🧩 Detailed BRD: Demand Forecasting & Order Suggestions

Overview

The Demand Forecasting & Order Suggestions module is the intelligent core of SMART-STOCK, designed to eliminate the guesswork from hardware store inventory management. By automatically analyzing historical sales data processed through the POS, this module predicts future product demand. It alerts the Admin before items run out and provides mathematically backed recommendations on exactly what to restock and in what quantities, ensuring capital is not tied up in slow-moving stock while preventing stockouts of highly requested products.

Scope

- Continuous background analysis of daily sales velocity per item
- Calculation of reorder points based on historical demand
- Generation of a "Suggested Orders" dashboard for the Admin
- Displaying recommended restock quantities to return inventory to optimal levels
- Ability for the Admin to dismiss a suggestion or mark it as "Ordered"
- Exporting the suggested order list (e.g., to CSV/PDF) to facilitate physical purchasing

Out-of-Scope

- Direct integration with external hardware supplier APIs
- Automated creation and emailing of Purchase Orders (POs) to vendors
- Tracking supplier lead times or delivery transit times dynamically
- Predicting seasonal spikes based on external factors (e.g., weather data or market trends)

Functional Requirements

Requirement	Actor	Priority	Rationale

The system shall calculate the Average Daily Sales (velocity) for every item based on the preceding 30 days of transaction data.	System	Must	Establishes the baseline metric for how fast a product is moving.
The system shall automatically flag an item for reorder when its Current Stock falls below its calculated 7-day demand threshold.	System	Should	Ensures the Admin is notified before the item physically runs out.
The system shall calculate and display a specific "Suggested Order Quantity" designed to restore the item's stock to a 30-day supply.	System	Should	Tells the store owner exactly how many units to buy, preventing over-stocking.

The system shall provide a dedicated "Order Suggestions" dashboard displaying the flagged items, their current stock, velocity, and suggested order quantity.	Admin	Must	Centralizes all purchasing decisions in one easy-to-read interface.
The system shall allow the user to mark a suggestion as "Ordered" or "Dismissed" to remove it from the active dashboard.	Admin	Must	Prevents the dashboard from becoming cluttered with items the owner has already addressed.
The system shall entirely hide this module and its underlying data from Staff-level accounts.	System	Must	Maintains role-based security boundaries for business intelligence data.
The system shall allow the user to export the active Order Suggestions list as a printable format or spreadsheet.	Admin	Should	Allows the owner to take the list offline when calling suppliers or visiting wholesale warehouses.

Non-Functional Requirements

Category	Requirement
Performance	Forecasting calculations shall run asynchronously (e.g., via a scheduled nightly job) to prevent UI lag during regular business hours.
Usability	The dashboard must clearly differentiate critically low items (e.g., under 3 days of supply) using distinct visual indicators (like red text/icons).
Reliability	If an item has less than 7 days of sales history, the system shall bypass the forecasting algorithm and rely on a static minimum threshold (e.g., 10 units).
Security	Any attempt to access the suggestion export endpoint without a valid Admin session token shall result in a 403 Forbidden response.

Constraints

- The accuracy of the forecast is strictly bound by the accuracy of the POS data; if staff fail to ring up items correctly, the suggestions will be skewed.
- Forecasts rely purely on linear historical data and cannot natively predict sudden, unprecedented spikes in demand.

Limitations

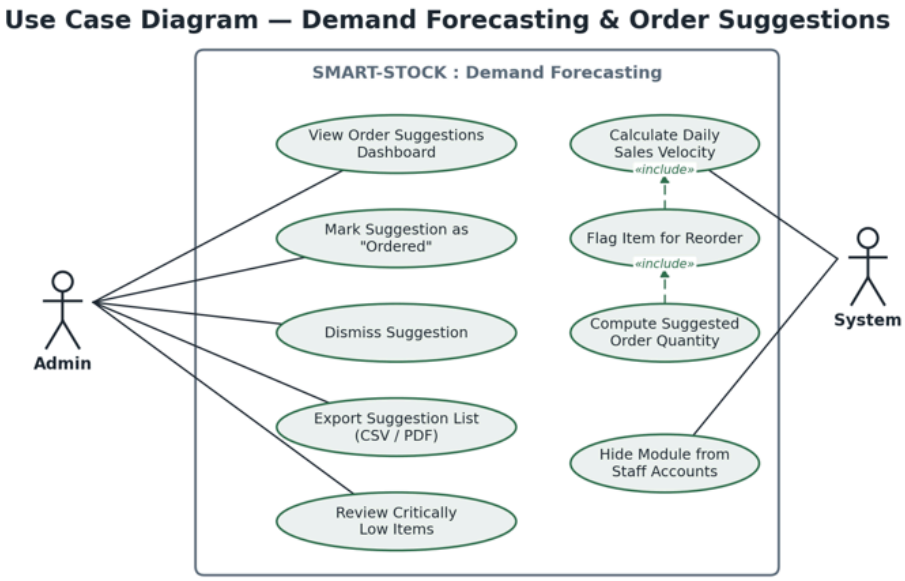
- Because the system does not track supplier lead times, the 7-day safety threshold is a static assumption that may not be sufficient for hard-to-source materials.

Assumptions

- The store owner reviews the dashboard regularly (e.g., weekly) to place orders before the 7-day safety stock depletes.
- The business prefers to maintain approximately one month (30 days) of stock on hand for standard items to balance storage space and availability.

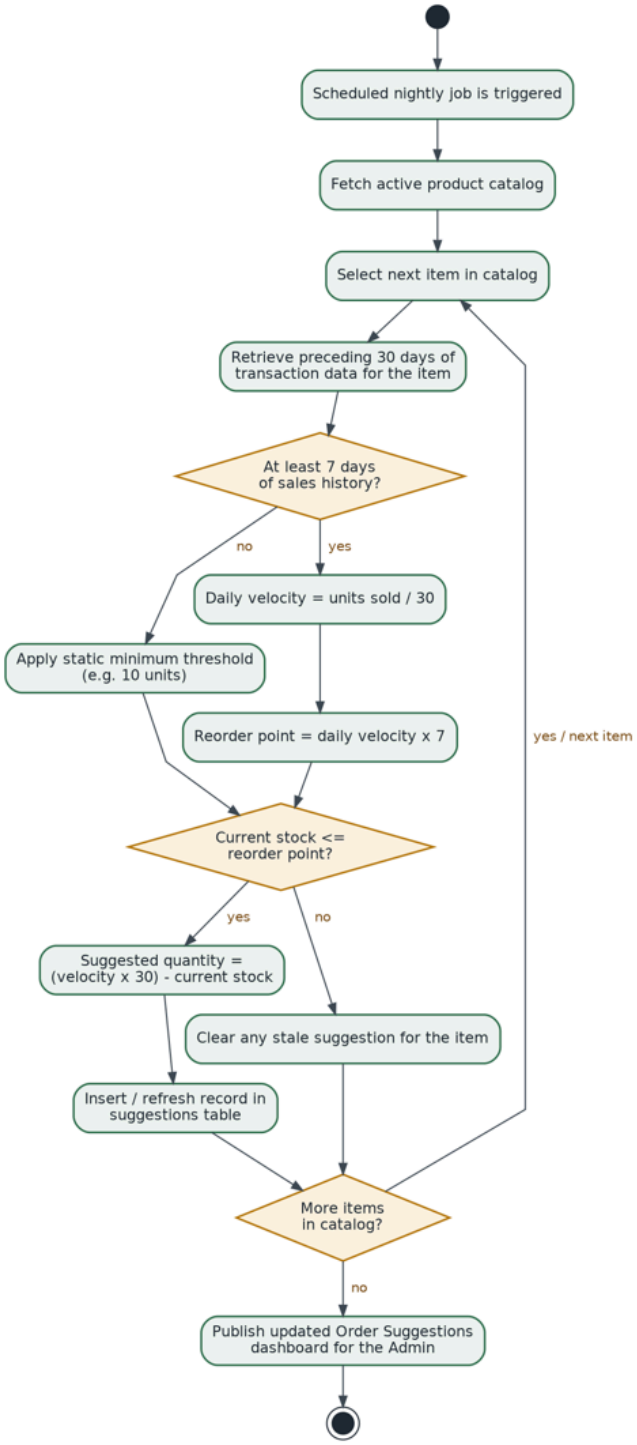
Requirements Model

- **Use Case Diagram**



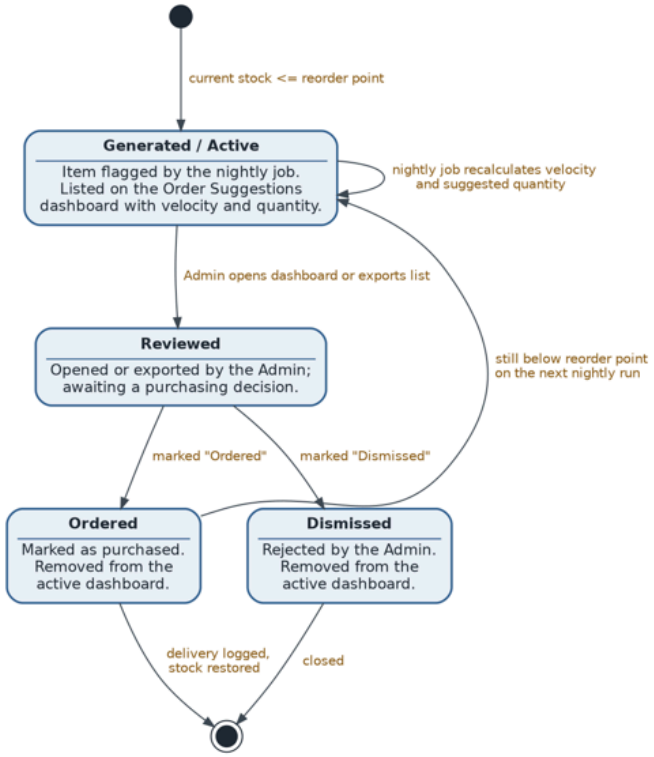
- **Business rules**
  - Daily Velocity = Total Units Sold / 30 days.
  - Reorder Point (Threshold) = Daily Velocity * 7 days.
  - Suggested Order Quantity = (Daily Velocity * 30 days) - Current Stock.
  - If Current Stock <= Reorder Point ➔ Add to Suggestions List.
- **Activity Diagram**
  - *Nightly Forecasting Job Workflow*

Activity Diagram — Nightly Forecasting Job (Demand Forecasting)



- **State model**
  - *Suggestion Record Lifecycle*

State Model — Suggestion Record Lifecycle (Demand Forecasting)



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